Matthieu Broisin

ROBOTICS & EMBEDDED SYSTEMS ENGINEER

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PROFILE

Engineer with more than five years of experience across applied research and industry, specialising in the development, integration, and maintenance of technical systems. My work experiences span robotic systems, embedded technologies, control and vision systems, and industrial precision motion systems. My multidisciplinary background allowed me to contribute to the development of novel robotic solutions for research and to investigate and solve complex technical issues in industrial applications.

PROFESSIONAL EXPERIENCE

Support Engineer — ETEL · Môtiers, NE · 2025–Present

Troubleshooting and testing of precision motion systems for the semiconductor industry, investigating magnetic, electrical, mechanical, firmware and software issues. Analysis, tests and statistics of returned units. Customer training and development of customized software solutions.

SOFTWARE
C - C++ - C# - Visual Basic – Python – Bash – Control theory
HARDWARE
Power electronics – Direct drive linear and torque motors – Encoder technologies – Positioning systems
TESTS & DIAGNOSTICS
Vibration analysis – Statistics – Laser interferometry (Renishaw) – Troubleshooting – Vacuum and pressure – Mass continuity

Research Engineer — EPFL, Mobile Robotics Group (MOBOTS) · Lausanne, VD · 2019–2024

Mechanical, electrical, firmware and high-level software design of a robotic beekeeping frame for Animal-Robot interaction research, able to modulate honeybees' foraging behavior.

SOFTWARE
C - C++ - Python – Bash
HARDWARE
Control electronics – Analog electronics – PCB design and assembly – Piezo-electric devices
METHODS
Signal processing – Image and video processing – Experimental design – Real time data processing – Biohybrid closed loop systems

COMPETENCIES

Technical

Signal/Image Processing
Electronical & Mechanical Design
Control Theory
Data Analysis/Synthesis
Backend Development
Firmware Development
Statistics

Transversal

Pedagogy
Project management
Academic writing
Information synthesis
Presentations/Defenses
Problem solving
Customer service

Tools

Microsoft Suite	Adobe Suite
Solidworks	Catia
Altium designer	VS Code
Github (CLI & GUI)	mySQL
InfluxDB	Synology Suite
Android Studio	Latex
Jira	STMCube

EARLIER WORK & QUALIFICATIONS

Research Intern — EPFL, Biomedical Imaging Group (BIG) · Lausanne, VD · 2019

Modelling and evaluation of Light-based Computed Tomography reconstruction algorithms.

Master's Thesis — Broad Institute of MIT and Harvard · Cambridge, MA · 2018–2019

End-to-end Deep-Learning framework design for automatic nucleus segmentation in electron microscopy images.

Engineering Internship — Bobst SA) · Mex, VD · 2018

Software tools development for selection of relevant images from a large dataset of production flaws in packaging industry.

BSc and MSc in Robotics — EPFL) · Lausanne, VD · 2013–2019

TEACHING & ACADEMICS

Mathematics Teacher – PrEP Lausanne · 2025

Academic Assistant — EPFL · 2019-2024
Coordination, course material preparation and grading of Basics of Mobile Robotics (Master) and Embedded Systems (Bachelor) classes.

Student Assistant — EPFL · 2019-2024
Helping students in the Signal Processing (Bachelor) and Image Processing (Master) classes.

Private Mathematics Teacher · 2010–2025

PUBLICATIONS & AWARD

Caicedo, J.C et al. · 2019
"Evaluation of deep learning strategies for nucleus segmentation in fluorescence images." Cytometry PartA

Ilgün, A. et al. · 2021
"Bio-hybrid systems for ecosystem level effects." In Artificial Life Conference Proceedings

<incoming> Broisin, M. et al. · 2027
"Integrated robotic agent for vibration-based interaction with honeybees."

STI Teaching Assistant Award · 2022
EPFL's School of engineering award for outstanding teaching contribution at EPFL.

LANGUAGES

French	English	Spanish
Native	C1	A2